5. PERFORMANCE DATA

5.1 OEI CLIMB PERFORMANCE

OEI climb performance charts (see Fig. 1 thru 6) are presented for OEI/MCP, 30-min and 2.5-min power ratings at different airspeeds (V_Y and V_{TOSS}). These charts show rate of climb and height gain data as functions of pressure or takeoff altitude, outside air temperature and gross mass.

For OEI operations requiring engine anti-icing ON, reduce rate-of–climb values by 200 fpm.

NOTE Chart data are based on clean configuration.

EXAMPLE: (see Fig. 1)

Determine: **OEI rate of climb**

Known:	Pressure altitude	6500	ft
	OAT	0	°C
	Gross mass	1840	kg
	Airspeed	50	KIAS (V_{TOSS})

Solution:
1. Enter appropriate chart at known pressure altitude (6500 ft).
2. Move horizontally right to known OAT (0 °C).
3. Move vertically down to the reference line and trace further down following parallel to the gross mass guide lines.
4. Reenter chart at known gross mass (1840 kg).
5. Move horizontally right to intersect tracing from above.
6. From point of intersection move vertically down and read rate of climb = 380 ft.

EXAMPLE: (see Fig. 5)

Determine: **OEI height gain over a horizontal distance of 100 ft**

Known:	Takeoff altitude	9700	ft
	OAT	15	°C
	Gross mass	1550	kg
	Headwind	15	kt
	Airspeed	50	KIAS (V_{TOSS})

Solution:
1. Enter appropriate chart at known takeoff altitude (9700 ft).
2. Move horizontally right to known OAT (15 °C).
3. Move vertically down to the reference line and trace further down following parallel to the gross mass guide lines..
4. Reenter chart at known gross mass (1550 kg).
5. Move horizontally right to intersect tracing from above.
6. From point of intersection move vertically down to reference line and trace further down following parallel to the headwind guide lines.
7. Reenter chart at known headwind (15 kt).
8. Move horizontally right to intersect tracing from above.
9. From point of intersection move vertically downwards and read height gain over a horizontal distance of 100 ft = 6.4 ft.

OEI RATE OF CLIMB - FLIGHT PATH SEGMENT I

1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE) ANTI-ICING – OFF

TAKEOFF SAFETY SPEED – 50 KIAS BLEED AIR CONSUMERS – OFF

NOTE WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

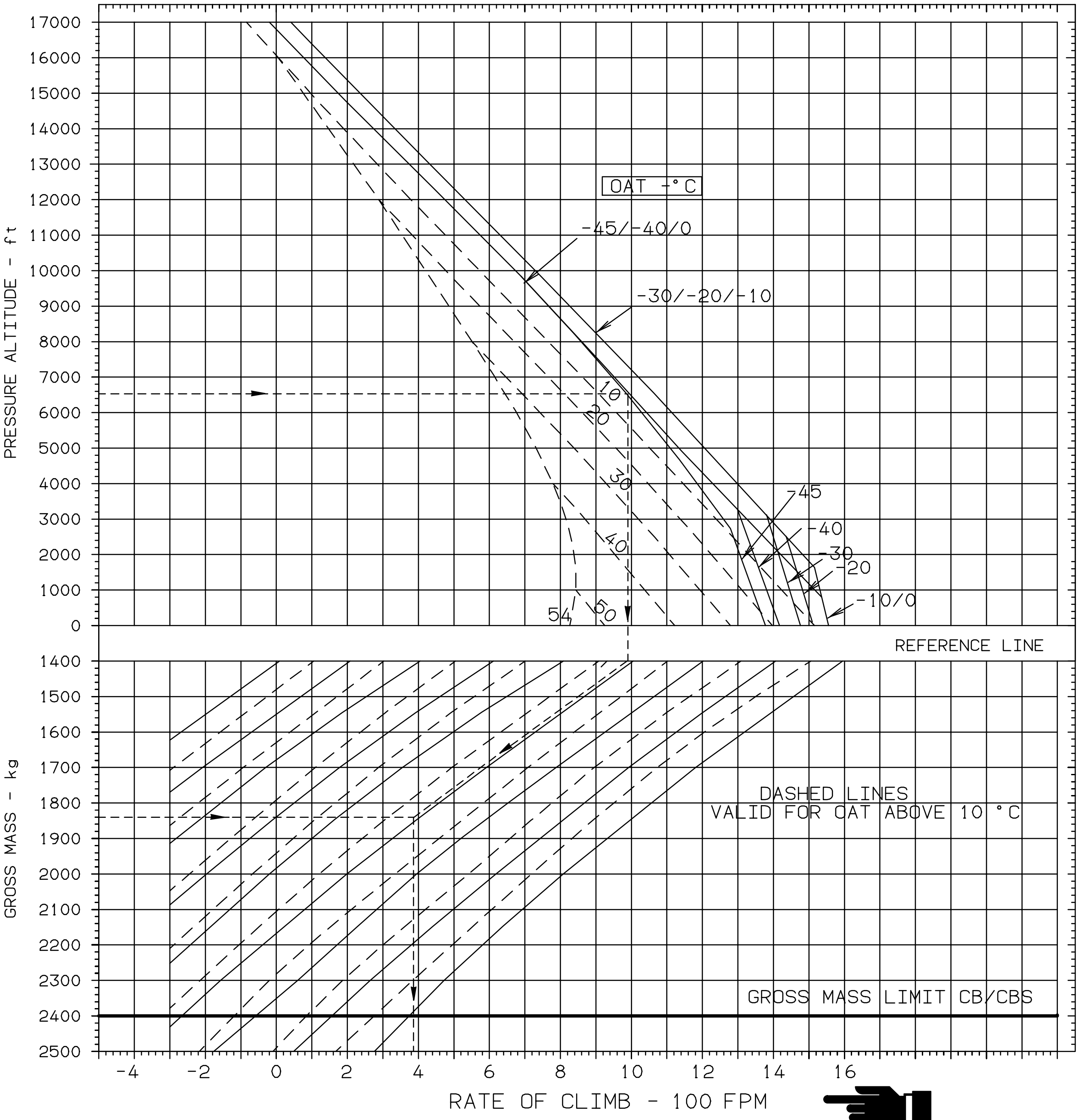

Fig. 1 OEI Rate of Climb - Flight Path Segment I

OEI RATE OF CLIMB - FLIGHT PATH SEGMENT II

1 X ALLISON 250-C20B

OEI – 30-MIN POWER (810 °C TOT, 105% TORQUE)
ANTI-ICING – OFF
BEST RATE-OF-CLIMB SPEED – 60 KIAS
BLEED AIR CONSUMERS – OFF

NOTE WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

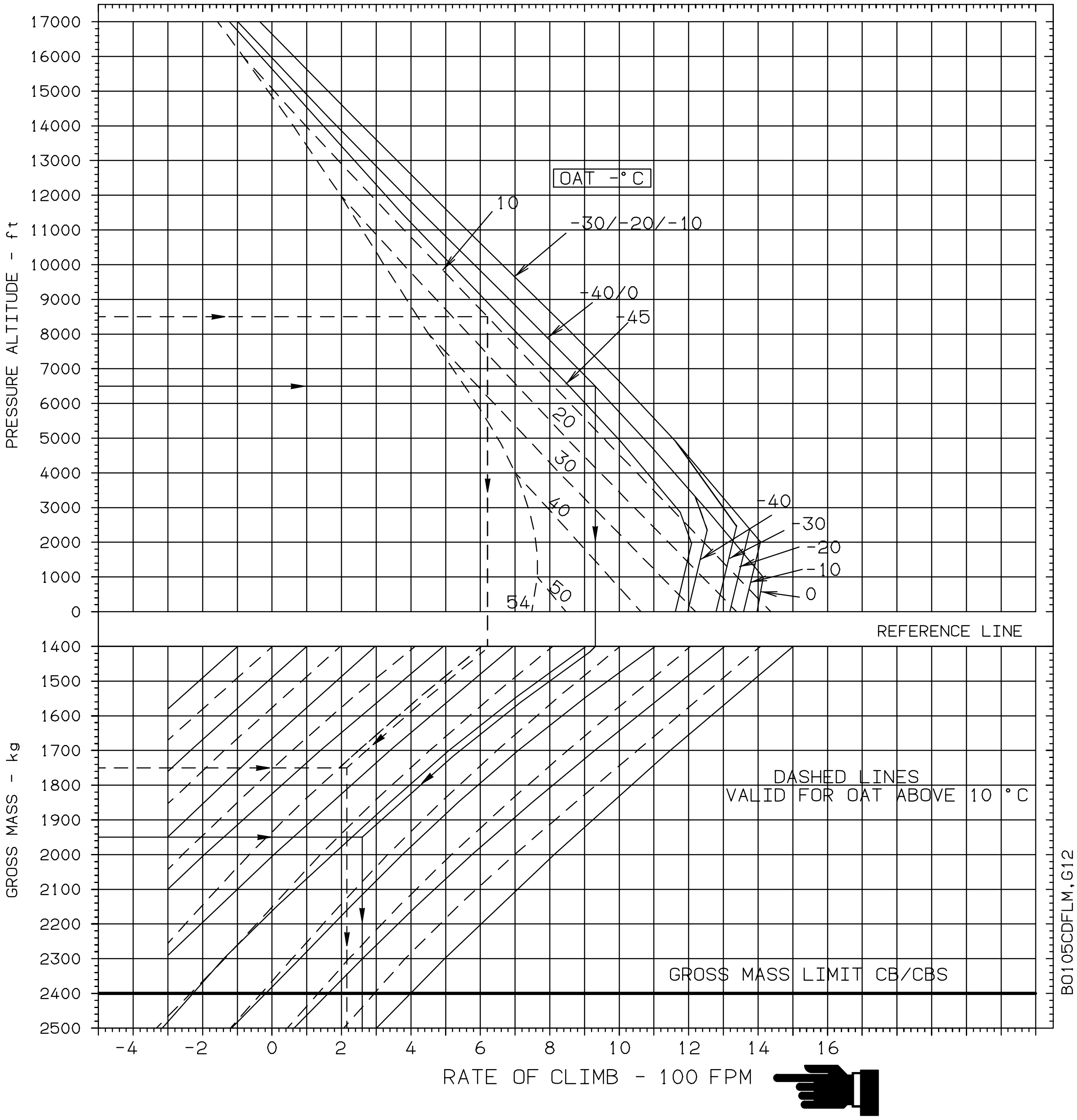

Fig. 2 OEI Rate of Climb - Flight Path Segment II

LBA APPROVED

OEI RATE OF CLIMB

1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE)
BEST RATE-OF-CLIMB SPEED – 60 KIAS

ANTI-ICING – OFF
BLEED AIR CONSUMERS – OFF

NOTE WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

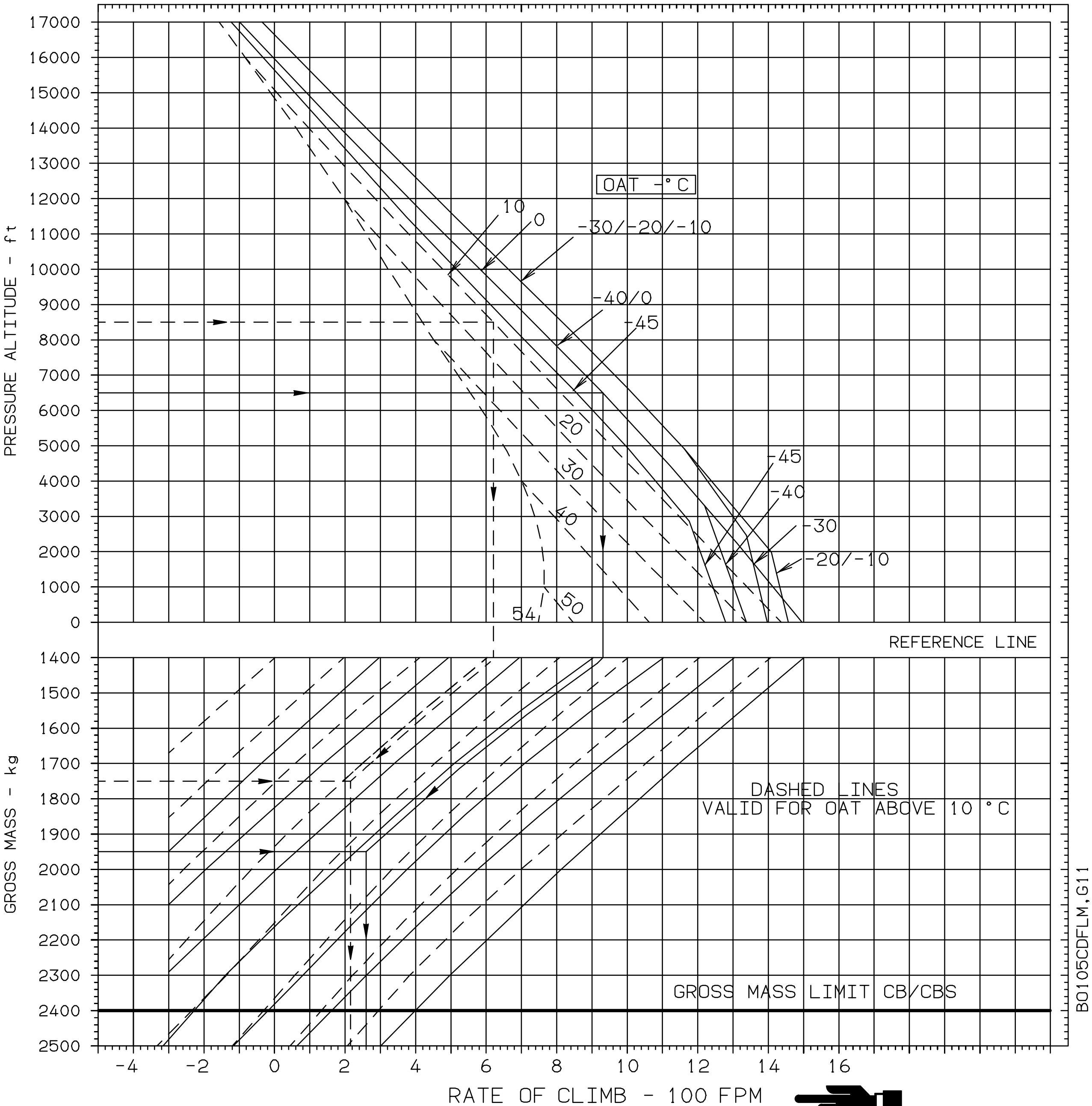

Fig. 3 OEI Rate of Climb - 2.5-min Power, V_Y

OEI RATE OF CLIMB

1 X ALLISON 250-C20B

OEI – MAX CONTINUOUS POWER (779 °C TOT, 105% TORQUE) ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS BLEED AIR CONSUMERS – OFF

NOTE WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

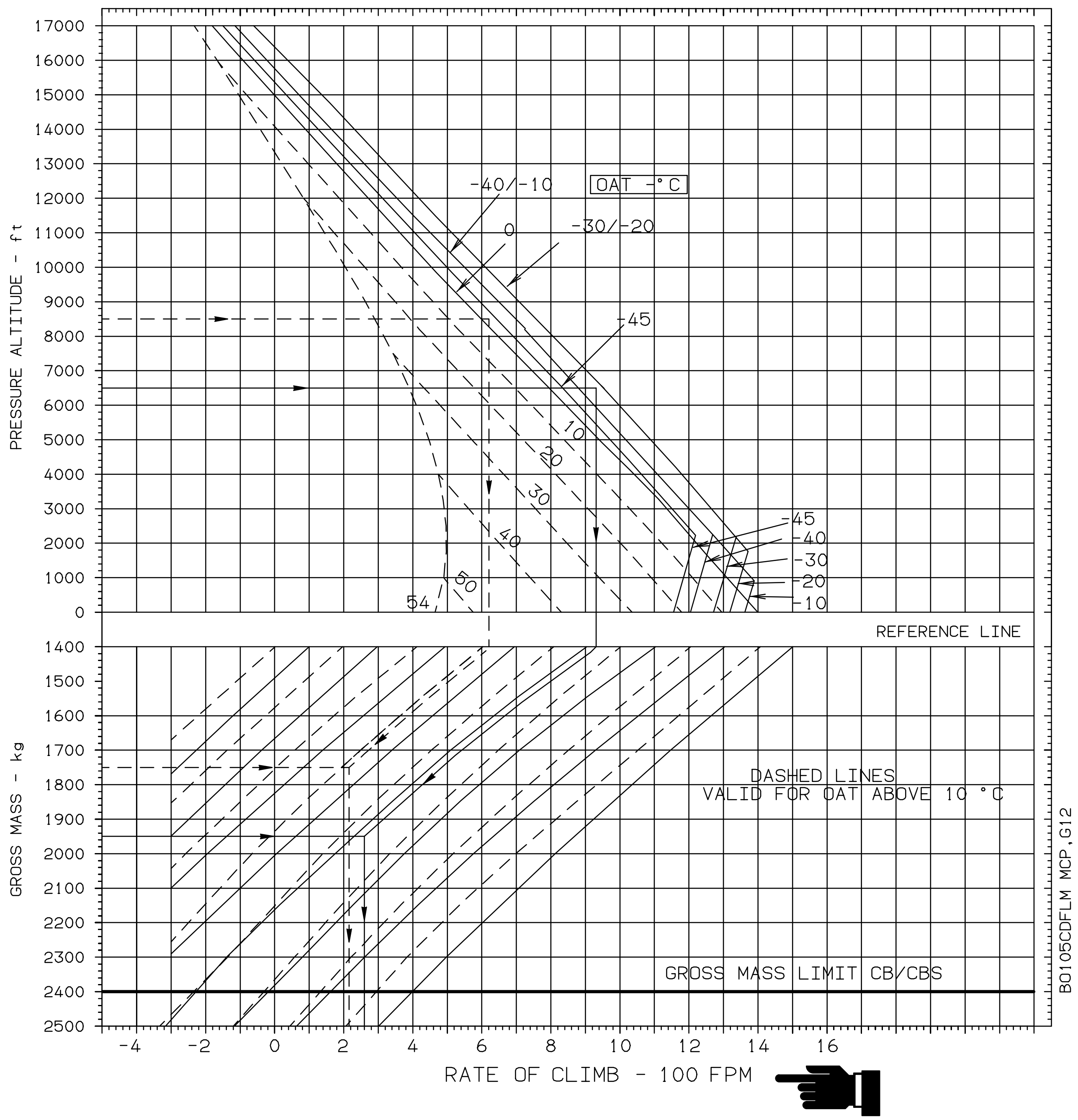

Fig. 4 OEI Rate of Climb - MCP, V_Y

OEI HEIGHT GAIN - FLIGHT PATH SEGMENT I

1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE) ANTI-ICING – OFF
TAKEOFF SAFETY SPEED – 50 KIAS BLEED AIR CONSUMERS – OFF

NOTE WITH ENGINE ANTI-ICING ON THE HEIGHT GAIN IS REDUCED BY 4FT/100FT.

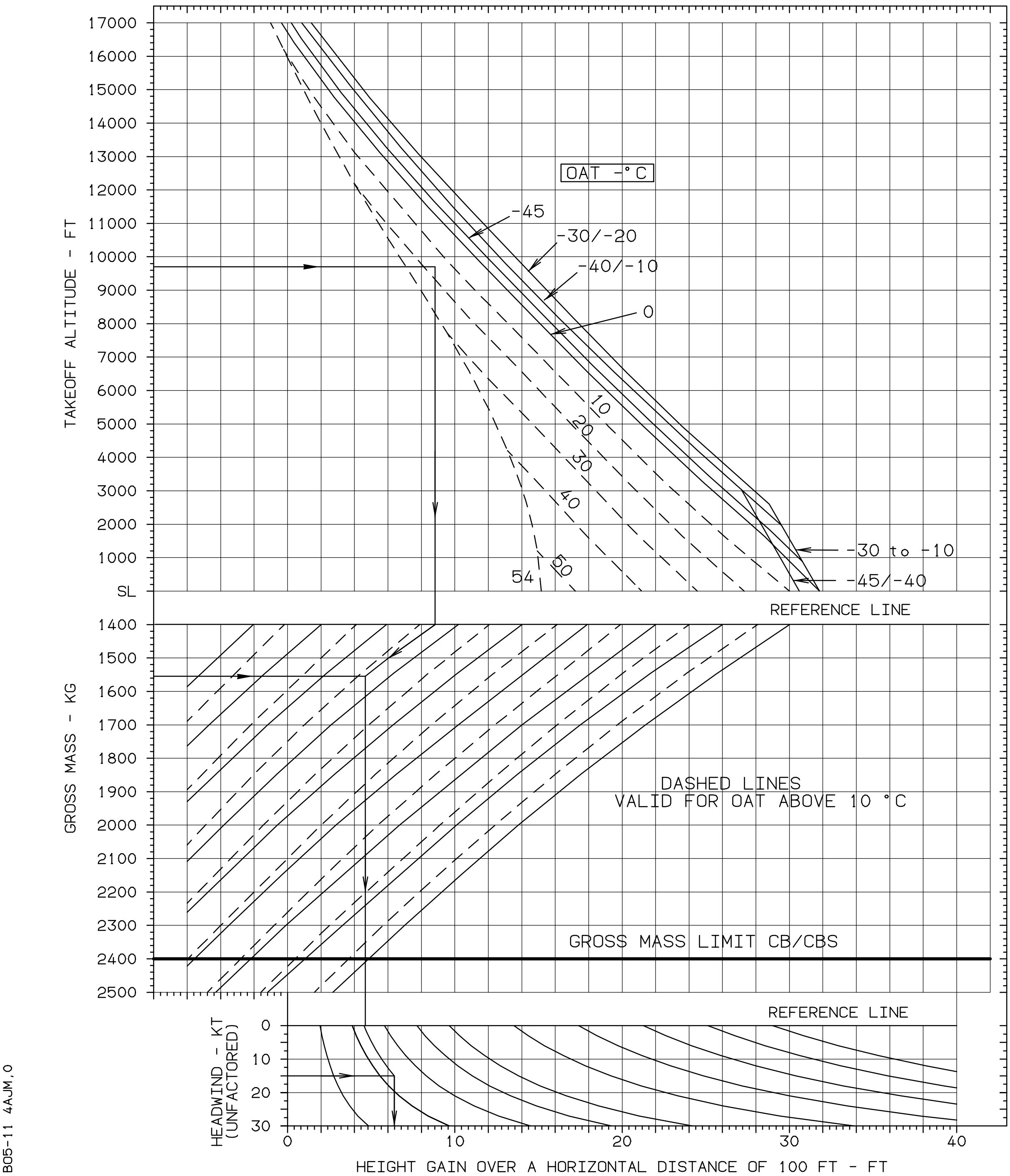

Fig. 5 OEI Height Gain - Flight Path Segment I

OEI HEIGHT GAIN - FLIGHT PATH SEGMENT II

1 X ALLISON 250-C20B

OEI – 30-MIN POWER (810 °C TOT, 105% TORQUE) ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS BLEED AIR CONSUMERS – OFF

NOTE WITH ENGINE ANTI-ICING ON THE HEIGHT GAIN IS REDUCED BY 3FT/100FT.

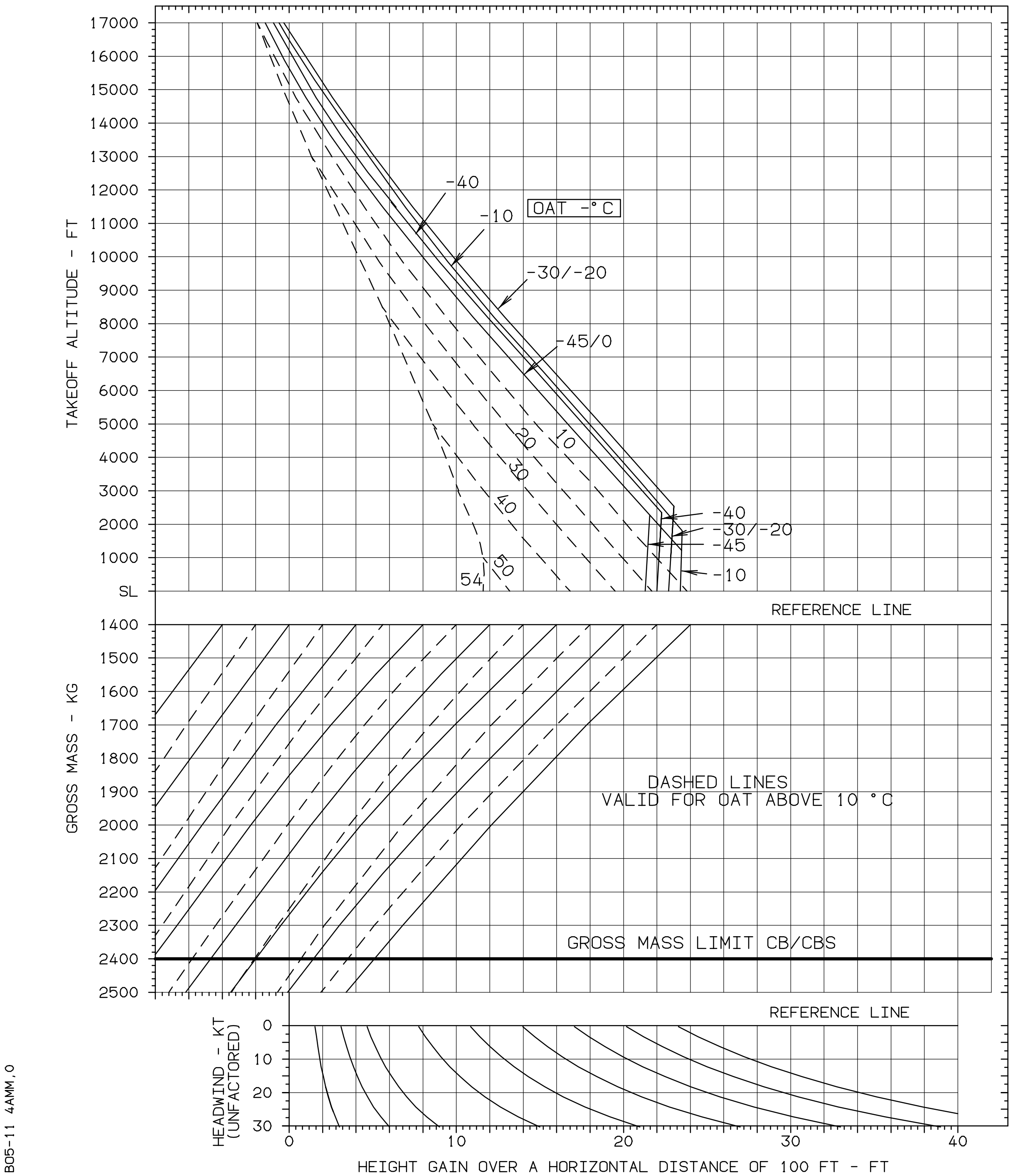

Fig. 6 OEI Height Gain - Flight Path Segment II